

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> First let's calculate how long it takes to download 40% of the file . Since the file size is 200 GB . 40% of 200 GB is $0.4 * 200 = 80$ GB . Given that she can download at a speed of 2 GB / minute . To download 80 GB would take $80 / 2 = 40$ minutes . Now let's consider the entire process . Initially she downloads 40 GB in 20 minutes . Then she has to restart her download from the beginning . To download the remaining 80 GB again would take $80 / 2 = 40$ minutes . Additionally she has to install updates which takes 20 minutes . Now let's calculate the total time . $20 + 40 + 40 + 20 = 120$ minutes . Therefore to download the entire file would take 120 minutes . </think><answer>120</answer>

